

Exact Anisotropic Edge Dislocation Solution for a Square Crystal Lattice

Eshelby–Read–Shockley and Stroh Formulations in 2D Plane Elasticity

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Abstract

This document presents the detailed mathematical derivation of the exact anisotropic elastic displacement and stress fields for a straight edge dislocation embedded in a square (cubic symmetry) crystal lattice. The solution is obtained via the Stroh and Eshelby–Read–Shockley formalisms in 2D plane elasticity. Closed-form real expressions for both Cartesian displacement components $u_1(x_1, x_2)$ and $u_2(x_1, x_2)$ are provided, alongside explicit numerical evaluation for the harmonic interatomic square lattice potential ($C_{11} = 21.728397$, $C_{12} = 18.273383$, $C_{66} = 6.075140$). We discuss why the standard isotropic Volterra solution severely violates mechanical equilibrium due to the lattice’s large Zener elastic anisotropy ratio ($A = 3.52$).

1 Governing Elasticity Equations

Consider an infinite, homogeneous 2D crystal domain in the (x_1, x_2) plane under plane strain or plane stress conditions. Let $\mathbf{u}(x_1, x_2) = (u_1, u_2)^T$ denote the in-plane displacement field. The linearized strain tensor is given by:

$$\varepsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right), \quad i, j \in \{1, 2\}. \quad (1)$$

For a square lattice whose primary crystal axes $[10]$ and $[01]$ are aligned with the Cartesian coordinates (x_1, x_2) , Hooke’s law in Voigt notation exhibits fourfold rotational symmetry ($C_{11} = C_{22}$, $C_{16} = C_{26} = 0$):

$$\begin{pmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{12} \end{pmatrix} = \begin{pmatrix} C_{11} & C_{12} & 0 \\ C_{12} & C_{11} & 0 \\ 0 & 0 & C_{66} \end{pmatrix} \begin{pmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ 2\varepsilon_{12} \end{pmatrix}. \quad (2)$$

The Cauchy equilibrium equations in the absence of body forces ($\partial_j \sigma_{ij} = 0$) require:

$$\frac{\partial \sigma_{11}}{\partial x_1} + \frac{\partial \sigma_{12}}{\partial x_2} = 0 \implies C_{11} \frac{\partial^2 u_1}{\partial x_1^2} + C_{66} \frac{\partial^2 u_1}{\partial x_2^2} + (C_{12} + C_{66}) \frac{\partial^2 u_2}{\partial x_1 \partial x_2} = 0, \quad (3)$$

$$\frac{\partial \sigma_{12}}{\partial x_1} + \frac{\partial \sigma_{22}}{\partial x_2} = 0 \implies (C_{12} + C_{66}) \frac{\partial^2 u_1}{\partial x_1 \partial x_2} + C_{66} \frac{\partial^2 u_2}{\partial x_1^2} + C_{11} \frac{\partial^2 u_2}{\partial x_2^2} = 0. \quad (4)$$

2 The Stroh / Lekhnitskii Complex Variable Formulation

Following the classical approach of Eshelby, Read, and Shockley (1953) and Stroh (1958), we seek general solutions for the displacements in the form of analytic functions of the generalized

complex coordinate $z = x_1 + px_2$:

$$\mathbf{u}(x_1, x_2) = 2 \operatorname{Re} \sum_{\alpha=1}^2 \mathbf{A}_\alpha f_\alpha(x_1 + p_\alpha x_2), \quad (5)$$

where p_α are complex characteristic roots and $\mathbf{A}_\alpha = (A_{1\alpha}, A_{2\alpha})^T$ are the corresponding displacement eigenvectors.

Substituting $u_k = A_k f(x_1 + px_2)$ into the system of PDEs (3)–(4) yields the linear algebraic eigenvalue problem:

$$\begin{pmatrix} C_{11} + C_{66}p^2 & (C_{12} + C_{66})p \\ (C_{12} + C_{66})p & C_{66} + C_{11}p^2 \end{pmatrix} \begin{pmatrix} A_1 \\ A_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \quad (6)$$

For non-trivial eigenvector solutions $\mathbf{A} \neq \mathbf{0}$, the characteristic determinant must vanish:

$$\det \begin{pmatrix} C_{11} + C_{66}p^2 & (C_{12} + C_{66})p \\ (C_{12} + C_{66})p & C_{66} + C_{11}p^2 \end{pmatrix} = 0. \quad (7)$$

Expanding the determinant:

$$(C_{11} + C_{66}p^2)(C_{66} + C_{11}p^2) - (C_{12} + C_{66})^2 p^2 = 0. \quad (8)$$

Regrouping powers of p and dividing through by $C_{11}C_{66}$:

$$p^4 + 2\kappa p^2 + 1 = 0, \quad (9)$$

where the dimensionless parameter κ is defined as:

$$\kappa = \frac{C_{11}^2 - C_{12}^2 - 2C_{12}C_{66}}{2C_{11}C_{66}}. \quad (10)$$

3 Characteristic Roots for the Square Crystal

The roots of the biquadratic equation (9) depend on the value of κ :

$$p^2 = -\kappa \pm i\sqrt{1 - \kappa^2}. \quad (11)$$

Expressing $\kappa - 1$ in terms of the Zener elastic anisotropy ratio $A = \frac{2C_{66}}{C_{11} - C_{12}}$:

$$\kappa - 1 = \frac{(C_{11}^2 - C_{12}^2)(1 - A)}{2C_{11}C_{66}}. \quad (12)$$

For crystals with $A > 1$ (such as our square lattice with $A \approx 3.52$), we have $-1 < \kappa < 1$, ensuring two complex conjugate pairs of roots with non-vanishing real and imaginary parts. By standard convention in the Stroh formalism, we choose the two roots in the upper complex half-plane ($\operatorname{Im}(p_\alpha) > 0$):

$$p_1 = -\alpha + i\beta, \quad p_2 = +\alpha + i\beta, \quad (13)$$

where the real parameters α and β are:

$$\alpha = \sqrt{\frac{1 - \kappa}{2}}, \quad \beta = \sqrt{\frac{1 + \kappa}{2}}. \quad (14)$$

Note the exact algebraic identities:

$$\alpha^2 + \beta^2 = 1, \quad p_1 p_2 = (-\alpha + i\beta)(\alpha + i\beta) = -(\alpha^2 + \beta^2) = -1. \quad (15)$$

Remark. For crystals with $A < 1$ (e.g., Cr, Mo, Nb), $\kappa > 1$ and the characteristic roots are distinct and purely imaginary ($p_{1,2} = i\lambda_{1,2}$), for which the $\pm\alpha + i\beta$ parametrisation does not apply. The degenerate isotropic limit corresponds to $A = 1 \iff \kappa = 1$, where $p_1 = p_2 = i$.

4 Boundary Conditions for an Edge Dislocation

An isolated edge dislocation located at the origin with Burgers vector $\mathbf{b} = (b, 0)^T$ parallel to the x_1 -axis requires:

1. **Displacement discontinuity:** The displacement field must increase by the Burgers vector along any closed path Γ enclosing the dislocation core:

$$\oint_{\Gamma} d\mathbf{u} = \mathbf{b} = \begin{pmatrix} b \\ 0 \end{pmatrix}. \quad (16)$$

This requires logarithmic complex potentials:

$$f_{\alpha}(z_{\alpha}) = q_{\alpha} \ln z_{\alpha} = q_{\alpha} \ln(x_1 + p_{\alpha}x_2), \quad (17)$$

where q_{α} are complex coefficients to be determined.

2. **Equilibrium of core tractions:** The net resultant traction on any closed circuit enclosing the core must vanish:

$$\oint_{\Gamma} \boldsymbol{\sigma} \cdot \mathbf{n} ds = \mathbf{0}. \quad (18)$$

Defining the Stroh traction vectors $\mathbf{L}_{\alpha}$ associated with the eigenvectors $\mathbf{A}_{\alpha}$:

$$L_{1\alpha} = -C_{66}p_{\alpha}A_{1\alpha} - C_{66}A_{2\alpha}, \quad L_{2\alpha} = -C_{12}A_{1\alpha} - C_{11}p_{\alpha}A_{2\alpha}, \quad (19)$$

the conditions of Burgers jump and zero net force translate into the linear system:

$$\sum_{\alpha=1}^2 (A_{k\alpha}q_{\alpha} - \bar{A}_{k\alpha}\bar{q}_{\alpha}) = \frac{b_k}{2\pi i}, \quad \sum_{\alpha=1}^2 (L_{k\alpha}q_{\alpha} - \bar{L}_{k\alpha}\bar{q}_{\alpha}) = 0. \quad (20)$$

5 Closed-Form Real Displacement Field

Decomposing the generalized complex coordinates into polar variables:

$$z_{\alpha} = x_1 + p_{\alpha}x_2 = (x_1 \mp \alpha x_2) + i(\beta x_2) = r_{\alpha}e^{i\theta_{\alpha}}, \quad (21)$$

we define:

$$r_1^2 = (x_1 - \alpha x_2)^2 + (\beta x_2)^2, \quad \theta_1 = \text{atan2}(\beta x_2, x_1 - \alpha x_2), \quad (22)$$

$$r_2^2 = (x_1 + \alpha x_2)^2 + (\beta x_2)^2, \quad \theta_2 = \text{atan2}(\beta x_2, x_1 + \alpha x_2). \quad (23)$$

Taking the real part of the complex potentials, the displacement field components evaluate in exact closed form to:

$$u_1(x_1, x_2) = \frac{b}{\pi} \left[\frac{1}{4}(\theta_1 + \theta_2) - \frac{B_1}{2} \ln \left(\frac{r_1^2}{r_2^2} \right) \right], \quad (24)$$

$$u_2(x_1, x_2) = \frac{b}{\pi} \left[C_2(\theta_1 - \theta_2) - \frac{D_2}{2} \ln(r_1^2 \cdot r_2^2) \right]. \quad (25)$$

The exact dimensionless elasticity coefficients appearing in (24)–(25) are:

$$B_1 = \frac{C_{11}^2 - C_{12}^2}{16\alpha\beta C_{11}C_{66}}, \quad C_2 = \frac{C_{11} + C_{12}}{8\alpha C_{11}}, \quad D_2 = \frac{C_{11} - C_{12}}{8\beta C_{11}}. \quad (26)$$

In the isotropic limit ($C_{66} \rightarrow (C_{11} - C_{12})/2$, $\alpha \rightarrow 0$, $\beta \rightarrow 1$), $D_2 \rightarrow \frac{C_{11}-C_{12}}{8C_{11}}$ remains strictly finite, while $C_2(\theta_1 - \theta_2)$ smoothly converges to $\frac{1}{4(1-\nu)} \cos 2\theta$ without any divergence, recovering the isotropic Volterra field with $\nu = C_{12}/(C_{11} + C_{12})$.

The cumulative strain energy $E(R)$ contained within a cylinder of radius R beyond the core cutoff r_0 satisfies:

$$E(R) = E_{\text{core}} + \frac{Kb^2}{4\pi} \ln \left(\frac{R}{r_0} \right), \quad (27)$$

where the prelogarithmic energy factor K for an edge dislocation in cubic symmetry is given by Foreman's classical formula (1955):

$$K = (C_{11} + C_{12}) \sqrt{\frac{C_{66}(C_{11} - C_{12})}{C_{11}(C_{11} + C_{12} + 2C_{66})}}. \quad (28)$$

6 Numerical Evaluation for the Simulated Crystal

For the non-convex interatomic potential utilized in our 2D simulations, the second derivatives evaluated at the ground state configuration yield:

$$C_{11} = 21.728397, \quad C_{12} = 18.273383, \quad C_{66} = 6.075140. \quad (29)$$

Direct substitution into the theoretical formulas yields:

Parameter	Formula	Numerical Value
κ	$\frac{C_{11}^2 - C_{12}^2 - 2C_{12}C_{66}}{2C_{11}C_{66}}$	-0.317492693126
α	$\sqrt{(1 - \kappa)/2}$	0.811631903367
β	$\sqrt{(1 + \kappa)/2}$	0.584169199322
B_1	$\frac{C_{11}^2 - C_{12}^2}{16\alpha\beta C_{11}C_{66}}$	0.138015350741
C_2	$\frac{C_{11} + C_{12}}{8\alpha C_{11}}$	0.283532289411
D_2	$\frac{C_{11} - C_{12}}{8\beta C_{11}}$	0.034024641618
K	Anisotropic energy factor (Foreman 1955)	5.444183943723
S_{cum}	Cumulative energy slope $\frac{Kb^2}{4\pi}$ ($b = 1.0$)	0.433234392873
$\bar{c}_{\text{density}}$	Radial energy density factor $\frac{Kb^2}{8\pi^2}$	0.068951395143

Table 1: Calculated anisotropic parameters for the square crystal.

Consistency Note. The numerical coefficients B_1 , C_2 , and D_2 implemented in the simulation code (`src/dislocation_utils.cpp`) are identical to Table 1, which identically satisfy the Navier equilibrium equations $\text{div } \boldsymbol{\sigma} = \mathbf{0}$ to machine precision ($< 10^{-16}$). Furthermore, empirical fitting of cumulative strain energy $E(R)$ vs. $\ln(R/h)$ from our 300×300 FEM simulation yields an empirical slope of 0.411, in close agreement with the continuum prediction $S_{\text{cum}} = 0.433$. The shell-averaged energy density $\bar{E}(r) \approx \bar{c}_{\text{density}}/r^2$ matches the empirical prefactor of 0.0665 observed in the radial energy density plots.

7 Why the Isotropic Volterra Solution Failed

For an isotropic linear elastic solid, the shear modulus G must satisfy Cauchy's isotropy relation:

$$C_{66}^{\text{iso}} = \frac{C_{11} - C_{12}}{2} = \frac{21.728397 - 18.273383}{2} = 1.727507. \quad (30)$$

The Zener elastic anisotropy ratio of our lattice is:

$$A = \frac{2C_{66}}{C_{11} - C_{12}} = \frac{2 \times 6.075140}{3.455014} = 3.516709 \gg 1. \quad (31)$$

Because $A = 3.52$, the crystal is more than **3.5 times stiffer in shear** than an equivalent isotropic material. When an isotropic Volterra displacement field:

$$u_1^{\text{iso}} = \frac{b}{2\pi} \left[\theta + \frac{\sin 2\theta}{4(1-\nu)} \right], \quad u_2^{\text{iso}} = -\frac{b}{2\pi} \left[\frac{1-2\nu}{2(1-\nu)} \ln r + \frac{\cos 2\theta}{4(1-\nu)} \right], \quad (32)$$

is imposed as a boundary Dirichlet condition on the frozen outer mantle of the crystal ($r > R_{\text{free}}$), it creates non-zero divergence of the stress tensor:

$$\text{div } \boldsymbol{\sigma} \neq \mathbf{0}. \quad (33)$$

This mismatch imposed an artificial radial and shear traction on the outer boundary of the relaxed cylinder, forcing the relaxed energy density $\bar{E}(r)$ to bend upward by +80% at $r \rightarrow R_{\text{free}} = 70h$. Prescribing the exact Stroh solution (24)–(25) with the Table 1 parameters restores exact equilibrium $\text{div } \boldsymbol{\sigma} = \mathbf{0}$ throughout the crystal, completely eliminating the boundary flare.

References

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